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REVIEW

Effect of Aeration Applied During Different Phases of Anaerobic Digestion

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Abstract Aerobic treatment has been investigated as a method to enhance putrescible substrate degradation and biogas production through anaerobic digestion (AD). A series of aeration methods has been studied in different phases of anaerobic digestion (before, during, or at a late stage of AD). Several research groups have applied aeration together with anaerobic digestion to improve hydrolysis and increase substrate conversion efficiencies. Aeration has been proven to reduce volatile fatty acids (VFA) accumulation during AD, reducing pH inhibition for methanogens, and thus increasing process yields. Aeration may represent an effective method to reduce substrates' toxicity (e.g. sulphur compounds), particularly when digestate, resulting from their anaerobic digestion, is destined for use on the land. However, a potential drawback is represented by decreased methane production observed as a consequence of excessive soluble COD consumption prior to the AD phase. Duration and intensity of aeration, substrate type, aeration method, temperature during aeration, and air application phase are deemed important factors capable of affecting the efficiency of this treatment. The present review aims to provide a comprehensive insight into research studies performed over the past decades to test the combination of aerobic treatment and anaerobic digestion of organic substrates.

Keywords Aeration treatment · Anaerobic digestion · Putrescible substrates · Hydrolysis yield · Waste stabilization · Bioenergy production

Introduction

To address the number of problems which characterize the anaerobic digestion (AD) process and to simplify the management of the final residue consisting of undigested organic compounds and humic substances, methods including the optimization of AD parameters (pH and substrate C/N adjustment, solid retention time prolongation, inhibitors removal, substrate pre-treatment, co-digestion) may be applied.

With the aim of ensuring a balanced environment for anaerobic archaea growth and better performance, enhancing biogas production and, concurrently, reducing the amount of final residues, numerous studies have been performed to develop strategies focused both on improving process yields and testing effective pre-treatments to incentivize hydrolysis (first crucial and rate limiting step for the degradation of organic substances) [1].

As reported by Ariunbaatar et al. [2], mechanical, thermal, chemical and biological pre-treatment methods may be applied. To focus on the latter, biological pre-treatments may be of an aerobic or anaerobic (two-stage AD) nature, or envisage the addition of specific enzymes such as peptidase, carbohydrase and lipase to the AD system [2–4].

Aerobic treatment appears to represent a promising option, in spite of the fact that oxygen has long been considered an inhibitor of methanogenesis. On the one hand, aerobic treatment may threaten the subsequent AD process, resulting in the consumption of part of the substrate and

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inhibiting anaerobic archaea. However, as long as the process is not too protracted, it may elicit the positive effect of merely removing the excess readily-biodegradable organic compounds, which commonly result in acidification due to accumulation of VFA inside the bioreactors and consequent drop in pH. Several studies have indicated that implementation of an aeration step prior to AD may reduce VFA accumulation in anaerobic digesters by means of oxidization, subsequently resulting in enhanced methane yields [5, 6]. On the other hand, however, aerobic treatment may also enhance hydrolysis of AD, stimulating enzyme activity and hydrolysing bacteria. A large group of facultative microorganisms underlying the hydrolysis of organic waste grow both in the presence of oxygen and under anaerobic conditions. Aerobic pre-treatment is an efficient method of increasing microbial growth of hydrolytic bacteria, and therefore enhancing production of extracellular enzymes, that catalyzes the hydrolysis of organic substrates [7].

Aerobic treatment has also been found to exert positive effects on methanogenic activity [8, 9]. Aeration may enhance AD performance in a number of ways according to the different modes of aeration employed. Indeed, aeration may improve hydrolysis, reduce the peak of VFA yielded by readily-biodegradable organics, thus contributing towards controlling pH throughout the AD process. The present paper undertakes a systematic review of the available data relating to the modalities and effects of aeration discussed in scientific literature for the purpose of AD process improvement.

Different Ways to Combine Aeration and Anaerobic Digestion

Aeration can be applied either as a pre-treatment or during the anaerobic digestion process; it may however also be performed solely in the latter stages of AD to enable a further production of methane and to improve the stability of digestate prior its successive treatment for reuse or disposal (see Fig. 1).

A series of different substrates, such as organic fraction of municipal solid waste (OFMSW), food waste from agricultural and industrial activities, sewage sludge, corn straw, grass silage, have been tested, with the scale of experimentation varying from batch to semi-continuous. Aeration modalities (natural aeration, air or oxygen injection, electrolytic aeration) varied according to the test scale and the phase of application.

Aerobic Treatment Prior to AD Process

Several studies have focused on the use of aeration as a pre-treatment method to improve the biodegradation and treatability of organic substrate [10–15]. The investigated substrates included corn straw, sewage sludge, OFMSW, vegetable wastes, horse dung, sisal pulp and olive mill wastewater. The majority of the experimentations highlighted the improvement of the first AD step, hydrolysis, as a consequence of pre-aeration [10, 11, 14, 16].

Tested aeration modalities were performed using air stones [16], air pumps [12, 14, 17], compressed air [10, 18], and air or oxygen injection [11, 19, 20]. A less intense but simpler aeration method was also achieved by leaving the tested BMP bottles open with or without shaking to maintain semi-aerobic conditions [15, 21]. Optimum aeration time varied from 9 to 96 h according to the different substrates and processes used [15, 18, 21]. With regard to batch test conditions, optimum oxygen loading ranged from 5 mlO_2/gVS to 10 mlO_2/gVS when oxygen injection was applied [11, 19]. When CSTR reactors were run with semi-continuous oxygen or air introduction, oxygen supply rates varied from 37.5 $mlO_2/L_R/d$ to 16 $LO_2/L_R/d$ [7, 17].

A summary of studies conducted using aerobic pre-treatment prior to AD process is provided in Table 1.

Fdez.-Güelfo et al. [22] compared aerobic pre-treatment using a series of different inoculums: mature compost, fungus and activated sludge; the results obtained revealed how that the use of compost yielded the best performance. Fdez.-Güelfo et al. [22–24] reported that mixing OFMSW with mature compost for 24 h resulted in a higher specific microbial growth rate (160–205% as compared to untreated OFMSW) and organic matter degradation, accompanied by a faster methane production. Optimum mature compost inoculation percentage was 2.5% (v/v), displaying

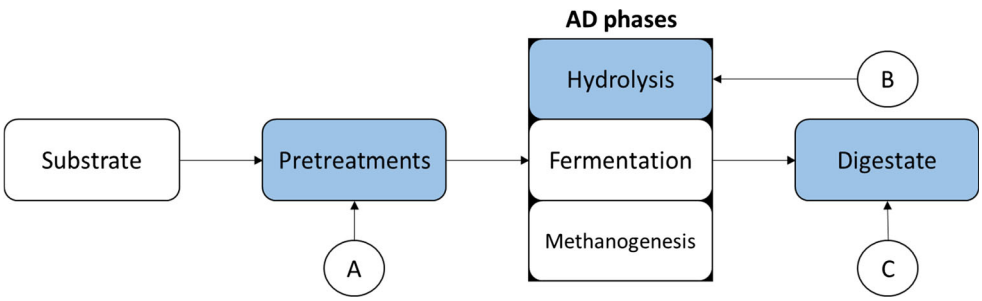


Fig. 1 Possible alternatives in the use of aeration within Anaerobic Digestion (both one and two-stage) operational schemes (A—pre-aeration; B—micro-aeration during AD process; C—aeration of digestate)

Table 1 Effect of aerobic treatment prior to AD process

Substrate	Aeration method and amount	AD process and test scale	Optimum results	Effects of pre-aeration	References
Corn straw	Oxygen injection under thermophilic conditions (0, 5, 10, 20, 30, 40 mlO_2/g VS)	Batch mesophilic BMP test	16.24% methane yield improvement (5 mlO_2/g VS)	Micro-aerobic condition caused higher hydrolysis	[11]
Digestate from primary fermentation of corn stover	Oxygen injection under thermophilic conditions (5, 10, 20 mlO_2/g VS)	Batch mesophilic BMP tests	28.45% methane yield improvement (10 mlO_2/g VS)	Not mentioned	[19]
Food waste and brown water	Oxygen injection (37.5 mlO_2/L_R/d; 4 days)	Batch mesophilic BMP tests	10% acetate increase in the hydrolysate and 23.0% methane yield improved (37.5 mlO_2/LR/d, 4 days)	The enhanced solubilisation of organics and control of VFA accumulation improved methane yield	[7]
Horse dung	Air injection at the bottom of the anaerobic digestion reactor (0.053 Lair/L_R/min; 48 h)	Mesophilic CSTR	18% methane yield decreased and no process kinetics enhanced	Excessive soluble COD consumption during the long aeration (48 h) led to a decreased methane production	[17]
OFMSW	Air injection (air was introduced into the sealed reactor until internal pressure was raised to a predetermined level)	DiCOM® reactor	Pre-aeration and wet thermophilic anaerobic digestion was able to stabilize OFMSW within a period of only 12 days	Increase of both cellulase and protease exoenzyme activities was observed	[10]
OFMSW	Air injection at the bottom of the anaerobic digestion reactor (0.0067 Lair/kg/min in; 2 h run/4 h stop)	Mesophilic CSTR	Biogas production doubled in AD reactors with micro-aeration	Better hydrolysis and acidification were observed	[12]
OFMSW	Partial composting as pre-aeration (20 °C 0.21 Lair/kg-OFMSW/min)	Batch mesophilic BMP tests	20% of VS was degraded during partial composting and 40% methane yield was lost	Start-up of the dry anaerobic batch digestion of OFMSW was accelerated	[28]
Olive mill wastewater	Air injection for 5 days (5 Lair/L_R/min)	Mesophilic CSTR	Methane production doubled from 0.16 m^3/kgCOD without pre-treatment to 0.39 m^3/kgCOD	Polyphenol concentration was reduced	[20]
Primary sludge and waste activated sludge	Natural aeration (bottles were left open to maintain semi-aerobic conditions) at 20 °C (12, 24, 48, 72, 96 h)	Batch mesophilic BMP test	VFA recovery (43 $gCOD_{VFA}$ kg^{-1} VS) and 14% methane potential improved (20 °C, 72 h)	Pre-aeration positive effect was linked to growth of fungi which have a strong biodegradation activity	[15]
Sewage sludge	Air injection (0.05 Lair/L_R/min; 0.5,1, 6, 24, 48, 96 h)	Batch mesophilic BMP tests	40% siloxane concentration decreased; 108% NH_3–N decreased; 25% methane yield improved (24 h)	SCODcr from sewage sludge microbial cell disruption increased	[16]
Sewage sludge	Air injection under thermophilic conditions (air rate 2.5 Lair/L_R/min; 1 day)	Batch mesophilic BMP tests	42% higher methane production rate and 15% higher TCOD removal (air rate 2.5 L/L_R/min, 1 days)	A greater diversity of bacteria and archaea populations was observed during mesophilic anaerobic digestion	[18]
Sewage sludge	Aerobic thermophilic pre-treatment by adding 1.2% VS of aerobic thermophilic sludge to methanogenic anaerobic sludge	Batch mesophilic BMP tests	Biogas production enhanced 2.2-fold	Aerobic thermophilic bacteria enhanced the production of biogas from anaerobically digested sewage sludge	[29]

Table 1 continued

Substrate	Aeration method and amount	AD process and test scale	Optimum results	Effects of pre-aeration	References
Sisal pulp	Natural aeration (digesters were left open to maintain semi-aerobic conditions and shaken at 135 rpm at 37 °C (0, 3, 6, 9, 12, 24, 48, 72 h)	Batch mesophilic BMP tests	The highest activity of hydrolytic enzymes was obtained at 9 h of pre-treatment	Higher activities of some extracellular hydrolytic enzymes	[21]
Vegetable and flower waste	Air injection (0.5 Lair/L_R/min, 5 min injection every 1, 4, 12, 24 h)	Batch mesophilic BMP tests	Hydrolysis efficiency with aeration treatment was 30.9–56.4% while hydrolysis efficiency was 57.7% under anaerobic conditions	Acidogenesis was promoted and accumulation of lactic acid was prevented	[14]

maximum hydrolytic activity with an increased COD of 51% and a 35.5% improvement in specific methane production [22–24]. Giordano et al. [25] applied aeration pre-treatment to prepare an inoculum for a two-stage hydrogen and methane production process. When food waste was used, a lower H_2 yield and a much higher CH_4 yield were obtained by the aerobic pre-treated inoculum compared to heat-shock, which suggested that aeration pre-treatment could preserve the hydrolytic activity and form a more structured microbial community [25–27].

Aeration Intensity and Aeration Methods

Aeration intensity or oxygen level should be given careful consideration during the pre-aeration stage due to the possibility of over-aeration reducing the methane potential of the digested substrate, and of too little oxygen potentially failing to produce any effect. A varied range of optimum aeration times, aeration intensity and aeration frequency have been adopted in lab-scale experiments due to the variety of substrates and AD processes. As shown in Table 1, when different aeration methods were used, the testing aeration time range varied considerably between 0 and 96 h. Ahn et al. [16] used air stones in the pre-aeration of sewage sludge. The results obtained showed an optimum aeration time of 24 h, resulting in a 25% methane yield improvement [16]. However, Peces et al. [15] also used primary sludge and waste activated sludge as substrates and left the bottles open to maintain semi-aerobic conditions. Their results demonstrated a 14% methane potential improvement following 72 h pre-fermentation [15]. From this comparison, the aeration time gap between the two studies could be attributed to the intensity of aeration. By simply leaving the substrate in contact with air it would be much more difficult for the oxygen to access the deeper parts of the medium.

Mshandate et al. [21] observed how 26% more methane was obtained after 9 h of aerobic pre-treatment (activated sludge mixed population as a source of inoculum in batch cultures) than with untreated sisal pulp waste. However, there was a loss of potential methane production with a longer pre-treatment of sisal pulp waste [21]. In this study moreover, aeration was performed simply by keeping bottles open and shaken, thus applying a more intensified aeration than that performed by Peces et al. [15]. The optimum aeration time was 9 h. Optimum aeration time may be linked not only to aeration intensity but also to substrate types. Other studies opted to apply intermittent aeration by injecting air (oxygen) or using controlled air pumps. Fu et al. [11] injected oxygen into the digester to pre-treat inoculated corn straw. The result showed a methane yield improvement of 16.24% under 5 mlO_2/gVS [11]. However, when these Authors used the same aeration method to treat digestate from primary digestion of corn straw [19] which was still containing a high amount of VS, the optimum oxygen level for obtaining the highest methane yield was 10 mlO_2/g VS; this suggests that more easily degradable substrate (raw corn straw) requires less oxygen compared to a more refractory substrate (digestate from corn straw).

Botheju et al. [30] reported how, under batch feed condition, oxygenation (0–16%) in anaerobic digestion produced no effect on volumetric biogas generation but lead to enhanced methane yield under the lowest oxygenation level of 1.3%. Increased oxygenation reduced CH_4/CO_2 ratio due to the significant increase in CO_2 production due to substrate oxidation [30].

Botheju and Bakke [31] have suggested that "electrolytic aeration" may represent an innovative concept for use in undertaking aeration pre-treatment. An electric current is used to introduce an aeration effect by means of water electrolysis, and electrolytically-generated hydrogen, together with oxygen, may also represent a substrate for AD. This concept has been tested in a study by Tartakovsky et al. [32], which demonstrated enhanced methane production (10–25%) from synthetic wastewater

in laboratory-scale anaerobic reactors equipped with electrodes for water electrolysis. The electrodes were installed in the reactor sludge bed and a voltage of 2.8–3.5 V was applied. Reactor stability was also improved in comparison to a conventional anaerobic reactor. Further, Tartakovsky et al. [33] reported a significant improvement of process performance and biogas quality (26% methane production improvement) by creating aerobic conditions in an electrolysis-enhanced anaerobic digestion process. In this study, cow manure and switchgrass were used as co-digestion substrates. Methane production from hydrolytic hydrogen and enhancement of hydrolysis of organic matter was used as a possible mechanism [33]. Chen et al. [34] applied direct voltage to create bio-stimulation, with the results obtained showing how methane production increased by 76.2% with an enhanced VS removal rate 26.6% with the applied voltage of 0.6 V. They also suggested that the positive effect might have been elicited by micro-aerobic conditions and from the available hydrogen resulting from water electrolysis. It is moreover worthy of mention that electricity consumption of 0.6 V can easily be compensated by the incremental energy producible from the higher recovery of methane [34]. Although the economic efficiency of AD reactors with electrolysis remains to be thoroughly evaluated, this electrical stimulation technology may constitute an innovative feasible method, and further studies should be carried out.

Effect of Aerobic Treatment on Hydrolysis

Aerobic treatment is capable of enhancing hydrolysis of AD, particularly when using lignocellulosic biomass and other refractory organic matters as a substrate.

The characteristics of organic substrates should be carefully evaluated prior to undertaking studies to investigate the impact of aeration pre-treatment. Physical and chemical characteristics of the substrate after aeration are largely dependent on the specific characteristics of the initial substrate. Pre-aeration of readily-biodegradable substrates, such as OFMSW, may reduce easily-degradable organics underlying the potential accumulation of VFAs [10] during the anaerobic process. In the case of not readily-biodegradable substrates, such as corn straw, pre-aeration improves degradability of the substrate [11]. A summary of pre-aeration effects focusing on hydrolysis performances for different substrates types is shown in Table 2.

Hydrolysis is catalysed by extracellular enzymes (such as protease, amylase, etc.) excreted by the facultative microorganisms. By investigating wet thermophilic anaerobic digestion of OFMSW through introduction of pressurized air, Charles et al. [10] reported improved hydrolysis during pre-aeration as a consequence of stronger cellulase and protease exo-enzyme activities. Zhu et al. [14] found that during hydrolysis of vegetable and flower wastes, cumulative protease and carboxymethyl cellulase activities increased with higher pre-aeration frequencies. Nonetheless, no increase of cumulative soluble organic matter is required as facultative bacteria consume VFA by means of aerobic respiration.

Pre-aeration with inoculum addition, resulting in biological hydrolysis, may prove to be more energy-saving and considerably more efficient than pre-aeration without inoculum. Lim and Wang [7] showed how inoculum addition influenced pre-aeration, as they observed 21 and 10% improvement in methane yields when pre-aerating inoculum-added substrates and pre-aerating substrates without inoculum, respectively. Accordingly, pre-aeration of substrate with inoculum is recommended in the absence of sufficient indigenous hydrolytic and acidogenic bacteria in the AD feedstock. Moreover, pre-aeration before AD may have indirect effects on hydrolysis, potentially enhancing biomass of facultative acidogenes and extracellular enzymes [31].

Pre-aeration Under Thermophilic Conditions

Performing pre-aeration at high temperatures may lead to a stronger synergistic effect. Jang et al. [18] evaluated the influence of thermophilic aerobic digestion as a sludge pre-treatment method on mesophilic anaerobic digestion. At the optimal conditions of 1-day thermophilic aerobic digestion and 20 days' mesophilic anaerobic digestion, the two-stage process yielded an approximately 42% higher methane production rate and 15% higher TCOD removal. This aerobic biological pre-treatment led to a higher diversity of the bacterial community in the anaerobic digester, resulting in enhanced hydrolysis and methane production [18].

Miah et al. [29] performed aerobic thermophilic pre-treatment prior to AD for enhanced methane production. They incubated sewage sludge at 65 °C for several months; by adding 1.2% VS of this aerobic thermophilic sludge to methanogenic anaerobic sludge they enhanced biogas production 2.2-fold [29].

Charles et al. [10] suggested that combined pre-aeration and wet thermophilic anaerobic digestion may be capable of stabilizing OFMSW over a 12-day period [10]. Fu et al. [11] investigated the effect of a combination of micro-aerobic and thermophilic (55 °C) pre-treatment on the anaerobic digestion of corn straw, with positive results (16.24% methane yield improvement). However, it cannot be concluded with any certainty that the hydrolysis process was accelerated under thermophilic conditions, as the study lacked control experiments at lower temperatures. However, high temperatures are also prone at times to

Table 2 Dynamics of physical and chemical characteristics of organic wastes when applying aeration

Substrate	Aeration method and amount	Indicators change		Reference
		Without aeration	With aeration	
OFMSW	Air injection (air was introduced into the sealed reactor until internal pressure was raised to a predetermined level)	VFA = 7.2 g/gTS	VFA = 1.8 g/gTS Aeration time: 3 days	[10]
Primary sludge	Natural aeration (bottles were left open)	Acetate = 0.92 gCOD/L	Acetate = 1.76 gCOD/L 20 °C semi-aerobic	[15]
Sewage sludge from a WWTP	Air injection (0.15 L/min; 0.5, 1, 6 24, 48, 96 h)	$SCOD_{Cr}$ = 2.65 g/L	$SCOD_{Cr}$ = 3.24 g/L Aeration time: 2 days	[16]
Vegetable and flower wastes	Air injection (0.7 L/min, 5 min injection every 1, 4, 12, 24 h)	–	TOC = 3397–4665 mg/L in the hydrolytic effluent Aeration time: 3 days	[14]

producing negative effects: Peces et al. [15] applied semi-aerobic fermentation to pre-treat primary sludge; under aerobic conditions, pre-fermentation at 37, 55, and 70 °C resulted in higher soluble COD, but elicited a 20% reduction in sludge methane potential at 55 and 70 °C [15]. Fdez-Guelfo et al. [22] combined thermochemical pre-treatments (air and sodium hydroxide) with biological pre-treatments (using mature compost, fungus *Aspergillus awamori* and activated sludge). Under aggressive temperature and pressure conditions, synthetic air employed as an oxidant agent may degrade a significant fraction of organic matter in OFMSW. However, amongst all assays the best result was obtained by thermochemical pre-treatment without air [22]. It may therefore be feasible to combine pre-aeration with other biological pre-treatments, such as the addition of compost, fungi or enzymes.

Degradation of Toxic Organics

Recent studies have revealed a reduction in the concentration of polyphenols, inhibitors of methanogenic archaea, and a doubling of methane yield following aerobic pre-treatment [20–22]. Gonzales–Gonzales and Cuadros [20] reported that 5 days of pre-aeration were enough to avoid polyphenols inhibition during the AD of olive mill wastewater. The mechanism of polyphenols reduction can be attributed to the oxidative action of laccases which require oxygen as a second substrate [21, 22]. Moreover, Wu et al. [13] reported how limited aeration was successful in aiding the removal of benzene, toluene, ethylbenzene and xylenes (BTEX), thus lessening the toxicity and enhancing degradability of petrochemical wastewaters. BTEX can be degraded in aerobic conditions since dioxygenases and monooxygenases enzymatic systems acts on metabolic pathways for their degradation [37, 38]. Further, Gavazza et al. [39] applied electrolysis to produce oxygen in order to achieve the aerobic degradation of aromatic amines in textile wastewaters. Simple aromatic amines can be mineralized under methanogenic conditions but sulfonated aromatic amines, on the other hand, are resistant and require specialized aerobic microbes for their mineralization [40]. Common substrates from solid wastes, such as food waste, corn straw and OFMSW did not usually raise the issue of the presence of toxic organics; however, when using sewage and industrial sludge means of reducing toxic or harmful substances should be considered. Ahn et al. [16] employed air stones for aeration to pre-treat sewage sludge prior to introduction into the AD process. The results obtained showed a 40% decrease in siloxane after 96 h' aeration because of its release into the atmosphere after the breakdown of extracellular polymeric substances (released by the disruption of microbial cells of sewage sludge) in which siloxane was adsorbed [16]. To date however, very few papers have been published that relate to the elimination of toxic organics during AD of organic wastes. Lastly, with regard to land use of digestate, the presence of toxic organics should be carefully evaluated before the application of pre-treatment during the AD process [41].

Assisting the Start-Up of AD Reactor

Pre-aeration has also been used to generate the heat required to meet AD energy requirements both under mesophilic and thermophilic conditions [10, 17]. Lagerkvist et al. [8] reported the efficiency of aeration in establishing methanogenic conditions in a full-scale AD reactor treating food waste. At start-up no methanogenic sludge

was available. The reactor was therefore fed with 1200 kg of total solids (TS) for 8 days and filled with thickened and slightly acidic sewage sludge. It was subsequently heated to 55 °C. Compressed air was subsequently blown into the digester and, within a month, a fully functional methanogenic culture was established. Charles et al. [10] found that pre-aeration of 48 h generated enough biological heat to increase the temperature of bulk OFMSW to 60 °C, sufficient for self-heating prior to start-up of thermophilic anaerobic digestion, without the need for an external heat source. Pre-aeration also reduced the excess of readily-degradable organic compounds in OFMSW, i.e. a common cause of acidification during start-up of the batch system [10]. Again, it is fundamental that over-aeration, which would consume the available substrate, should be carefully avoided, thus leaving the substrate available to methanogens for the production of biogas. Indeed, the results obtained showed how increased duration of aerobic pre-treatment limited the amount of methane gas formed rather than stimulating it. The highest cumulative CH_4 production was yielded by non-pre-aerated samples. Tenrummeler and Koster [28] applied an aerobic partial-composting step to solve the start-up unbalance of a dry anaerobic batch digestion of OFMSW, which resulted in the rapid degradation of easily-degradable organic compounds. The aerobic pre-treatment step lowered initial acid-formation rate and accelerated start-up of the dry anaerobic batch digestion of OFMSW. Nonetheless, partial composting caused a loss of 40% of potential methane yield [28]. As for the dry anaerobic digestion, pre-aeration or micro-aeration during the start-up of AD may reduce VFA accumulation whilst, at the same time, exerting no negative effect on methanogens due to the rapid oxygen consumption ability of facultative fermentative organisms, and the shielding effect of microbial aggregates [31].

Possible Drawbacks of Pre-aeration

Despite the several positive effects reported above, other researchers have referred to a clear drawback when performing aeration as pre-treatment prior to the AD process. According to the results obtained by Tenbrummeler and Koster [26], a partial pre-composting treatment of OFMSW resulted in a 19.5% VS loss. Aerobic pre-treatment resulted in a 10% lower methane production, likely due to the lengthy aeration period and excessive consumption of soluble COD prior to the anaerobic digestion phase [17].

Another important issue to consider is the additional operational cost due to pre-aeration. Literature provides only one study by Rafieenia et al. [42] mentioning that the cost and benefit analysis does not support the economic feasibility of the pre-treatment as the energy consumed for pre-aeration is much higher than the recovered one. Studies are still needed in order to investigate through a complete Life Cycle Assessment the affordability of pre-aeration.

Aeration Throughout the Entire AD Process

When applying aeration throughout the entire AD process in batch, CSTR, or UASB anaerobic bioreactors the investigated substrates used included sewage sludge, brown water, food waste, corn straw, cassava wastewater, grass silage, and evaporator condensate from a sulphite pulp mill [6, 8, 9, 15, 23, 30, 43–49]. The approaches implemented to aerobically stimulate anaerobic digestion performances included micro-aeration or limited oxygen injection throughout the entire AD process [9, 48–50]. Micro-aeration is defined as the introduction of small amounts of oxygen into an anaerobic biochemical process to enable both anaerobic and aerobic biological activities to occur within a single bioreactor [51]. When using micro-aeration or oxygen injection, the main purpose is to sustain the stability of AD reactors and to enhance the methane yield. Micro-aeration may be of use in reducing sulphides content and, therefore, minimizing the toxic effect of aqueous sulphides on acetogenic, methanogenic and micro-organisms [52–54]. Several studies have shown that limited oxygen may exert positive effects on hydrolysis [44, 48, 50].

A summary of studies that have used limited aeration during anaerobic digestion is reported in Table 3.

Aeration Intensity and Aeration Methods

Due to the limited solubility of oxygen, intermittent aeration was performed in all reviewed studies. When micro-aeration was performed during the acidogenic phase in a two-stage anaerobic digestion, 258 Lair/kgTS/d (54 LO_2/kgTS/d) was recommended for acidogenic leach bed reactor (LBR) treating food waste [46]. Jagadabhi et al. [50] suggested that micro-aeration at low flow rates (10 LO_2/kgVS/min) during the first 11 days of AD may improve VFA production without any significant increase in COD solubilisation. Similarly, in their two-phase CSTR, Lim et al. [44] added oxygen at a rate of 3 ml/min daily for 1–2 h after feeding. The results showed that low micro-aeration conditions (5 mlO_2/L_R/d) performed better than higher conditions (7 mlO_2/L_R/d) [44].

Studies performed to investigate single phase CSTR revealed that aeration intensity was lower than that needed for a two-phase CSTR [30, 47, 48]. Khongsumran et al. [48] supplied oxygen to a CSTR for 5 min every 2 h obtaining the highest cellulose hydrolysis efficiency by applying 3.0 mlO_2/L_R/d. Botheju et al. [30] exposed four CSTR reactors to different oxygenation levels performing daily air injection after daily substrate feeding. Only the

Table 3 Effect of limited aeration during the AD process

Substrate	AD process and test scale	Aeration method and amount	Results	Effects of pre-aeration	References
Brown water and food waste	Mesophilic two-phase CSTR	Oxygen injection to the liquid part of the reactor (5 $mlO_2/L_R/d$ and 7 $mlO_2/L_R/d$)	43% total VFA concentration increase with 5 $mlO_2/L_R/d$	Micro-aeration led to diverse bacterial communities *Firmicutes* enabled to metabolize a greater variety of substrates in the acidogenic reactor	[44]
Cassava wastewater	CSTR without temperature control	Oxygen injection into the CSTR (1.5 to 6.0 $mlO_2/L_R/d$)	Maximum hydrolysis efficiency: 62.57% for cellulose applying 3.0 $mlO_2/L_R/d$	Anaerobic hydrolysis was enhanced; Facultative bacteria increased resulting in higher population and more secreted enzyme	[48]
Cellulose	Batch mesophilic BMP tests	Oxygen injection (5 $mlO_2/L_R/d$)	No hydrolysis improvement, but shorter lag-phase time	Transformation of the cellulose to simple organic compounds in the early stages of AD was accelerated	[49]
Corn straw	Batch thermophilic BMP tests	Air injection (0, 12.5, 25, 50, and 100 $ml/L_R/d$)	16.5% methane yield improvement (12.5 $ml/L_R/d$)	Hydrolysis-associated microorganisms, *Firmicutes*, class *Clostridia* and order *Clostridiales* raised Oxytolerant *Methanosarcina* and doubled	[45]
Evaporator condensate from a sulphite pulp mill	Mesophilic UASB	Direct limited aeration in the UASB (aeration flow rate: 6 $ml/L_R/min$)	COD removal rate increase from 40% to 80%	Limited aeration caused no oxygen inhibition to the anaerobic microorganisms but instigated sulphide oxidization and H_2S removal	[6]
Grass-silage	Mesophilic leach-bed reactor	Air injection flow rate: 1.33 $L/L_R/min$ (for three minutes) on day 11 and 5.33 $L/L_R/min$ (for six minutes) on day 22	Fourfold increase in cumulative VFA production (for air flow of 1.33 $L/L_R/min$ on day 11)	The conversion of the produced leachate SCOD into VFA improved	[50]
OFMSW	Mesophilic CSTR	Air injection at the bottom of the anaerobic digestion reactor (1 L/min; 2 h run/4 h stop)	Reduction of DOC with an aeration time of 3 days	DOC reduced from 140 gC/kgTS (without aeration) to127 gC/kgTS (with aeration)	[12]
Sewage sludge	Mesophilic CSTR	Pure oxygen supply into the sludge recirculation 0.0033 $LO_2/L_R/min$	Better capability to deal successfully with overloads	The growth of hydrogenotrophic methanogens was promoted. This could favour acetic formation and help to maintain a stable pH	[9]
Sewage sludge	Mesophilic CSTR	Oxygen injection every 10 min into different O_2 dosing points with O_2 flow rate (0.005–0.034 $LO_2/L_R/d$)	Oxygen did not have a significant impact on digestion	O_2 transfer rate positively affected long-term microbial diversity	[43]
Sewage sludge and food waste	Thermophilic CSTR	Compressed air injection (41.7 mlair/kgTS/min)	Aerobic treatment may amend occasional acidification problems	The aerobic degradation of organic acids led to increased pH, thus promoting growth of methanogens	[8]
Synthetic food waste	Mesophilic two-phase UASB	Air pump at the bottom of the LBRs (0, 129, 258, and 387 Lair/kg TS/d)	5% COD faster leaching and 18% methane production enhancement with 258 Lair/kgTS/d	Micro-aeration could enhance acid fermentation and promote production of VFAs	[46]

Table 3 continued

Substrate	AD process and test scale	Aeration method and amount	Results	Effects of pre-aeration	References
Synthetic substrate (Starch 3.9, peptone 3.01 and yeast extract 3.58 g/L)	Mesophilic semi-CSTR	Daily air injection (0, 0.43, 0.86, 1.3 $mlO_2/L_R/d$)	30–55% methane yield increase	Initial VFAs accumulation was minimized	[30]
Synthetic substrate (Starch 3.9, peptone 3.01 and yeast extract 3.58 g/L)	Mesophilic semi-CSTR	Daily air injection after daily feeding (increasing oxygen loads of 0, 2.52, 5.04 and 10.07% daily feed COD)	Methane generation decreased linearly with the oxygen loading	Oxygen suppressed the formation of VFAs due to aerobic respiration	[47]
Synthetic wastewater	Mesophilic UASB	Oxygen supply through electrolysis (2.8–3.5 V)	10–25% methane production enhancement and reactor stability improvement	Hydrolysis and hydrogen sulphide oxidation were enhanced	[32]

lowest oxygenation level (3.25 $mlO_2/L_R/d$) produced a positive effect on methane production [30]. In a similar research study Botheju [47] observed a linear decrease in methane generation in the presence of an increased oxygen loading. In line with the above-mentioned studies, it may be feasible to conclude that the oxygen level should be around 3 $mlO_2/L_R/d$ in single CSTR systems [47].

Batch reactors were also used to test the feasibility of micro-aeration during the AD process. Fu et al. [45] achieved a 16.5% improvement in methane yield during the thermophilic AD of corn straw by injecting air into their bottles with a syringe (2.63 $mlO_2/L_R/d$). Diaz et al. [49], when testing cellulose batch AD, shortened the lag-phase time with daily oxygen supply (5 $mlO_2/L_R/d$). For refractory substrates such as corn straw or cellulose, aerobic and anaerobic treatment combination may not be sufficient.

Effect of Aerobic Treatment on Hydrolysis

Three modes of aerobic treatment (during the acidogenic phase in a two-stage anaerobic digestion process, in CSTRs, in batch anaerobic bio-digesters) were applied to evaluate the effects on hydrolysis during anaerobic digestion tests. Lim et al. [44] applied micro-aeration during the hydrolysis phase (acidogenic phase) in a two-stage anaerobic process performed to co-digest brown water and food waste. A 43% increase in total VFA concentration was achieved and the positive effect attributed to the growth and expansion of diverse bacterial communities promoted by micro-aeration [44]. Xu et al. [46] used synthetic food waste as a substrate and applied micro-aeration in a leach bed reactor (LBR). The results obtained showed that 2 h aeration enhanced carbohydrate and protein hydrolysis by 38 and 64%, respectively [46]. Jagadabhi et al. [50] also introduced air into an LBR and detected a quadruplicated VFA production, thus improving conversion of the produced leachate soluble COD (SCOD) into VFA. On the contrary, Lagerkvist et al. [8] introduced compressed air (41.7 mlair/min/kgTS) into their digester to achieve aerobic degradation of organic acids and to amend occasional acidification problems in a full scale AD plant. It was noted that aeration intensity should be chosen carefully, in accordance with the specific aim to be achieved, potentially improvement of hydrolysis or relieving of acid accumulation. Other studies provided aeration or injected air during the anaerobic process inside a CSTR. Khongsumran et al. [48] supplied oxygen into a CSTR treating cassava wastewater; maximum hydrolysis efficiency was achieved, namely 62.57%, under 3.0 $mlO_2/L_R/d$ due to the enhanced growth of facultative bacteria. Nonetheless, other studies reported no obvious hydrolysis improvement. Ramos and Fdz-Polanco [9] supplied pure oxygen during sludge recirculation and found that oxygen increased the ability to deal with overloads and enhanced digestion performances only under stressful conditions. Diaz et al. [49] applied micro-aeration at an oxygen supply of 5 $mlO_2/L_R/d$ in a batch mesophilic AD reactor treating cellulose. No improvement was detected in hydrolysis, although a shorter lag-phase time, which resulted from accelerating the transformation of the cellulose to simple organic compounds in the early stages of AD, was observed.

Effect of Aerobic Treatment on Oxidation of Hydrogen Sulphides

Limited oxygen injection into the anaerobic reactor was found to greatly reduce hydrogen sulphide accumulation

[49, 53, 55]. Oxygen is capable of oxidizing sulphides generated in anaerobic digesters in which S compounds are reduced to HS- and H_2S by the biological activity of anaerobic sulphate-reducing archaea (SRB). Sulphide content minimization enhances the entire AD process (see Table 4) by abating the toxic effect of aqueous sulphides on methanogenic and acetogenic organisms [49]. Other positive effects on AD performances reported in the literature are lag phase shortening, methane yield enhancement and biogas quality improvement [12, 49]. Micro-aeration may exert positive effects on the activity of anaerobic microorganisms. Other acknowledged explanations for the positive effects elicited under micro-aerobic conditions include a higher solubilisation of organic matter, intensified hydrolysis, and a more stable acidification [12]. Furthermore, micro-oxygenation may prevent hydraulic overloads by promoting the growth of both acetoclastic and hydrogenotrophic methanogenic archaea [9].

Possible Drawbacks

In this context of combined aerobic and anaerobic conditions during the AD process, a series of negative results have also been reported. Botheju et al. [47] concluded that total volumetric biogas generation was not significantly influenced by aeration achieved by means of daily oxygen injection in a 5.5 L bioreactor. A reduced methane yield was replaced by an increased CO_2 yield, and an increasingly negative effect on methane generation was observed on a par with the increasing aeration level [47].

Aeration in the Late AD Stage

As the end of the AD process approaches, increasingly lower quantities of organic matter can still be consumed to further produce methane. Nonetheless, digestate can be treated to further increase the substrate methane potential and improve the stability of digestate. It is worth to mention that digestate will, anyhow, need further post-treatment prior agronomic use or disposal. It was recently reported that by applying aeration during the late stages of AD, methane yield could be further improved [19]. In a study conducted by Fu et al. [19], digestate from primary fermentation of corn stover was treated under 55 °C with oxygen loads of 5, 10 and 20 ml/gVS (VS of residual substrate). The results obtained showed a 28.45% improvement in methane yield during secondary thermophilic micro-aerobic treatment with an optimal oxygen load of 10 ml/gVS. The Authors attributed this positive effect to the high hydrolysis rate under micro-aerobic thermophilic conditions. Indeed, the VS removal efficiency was 29.43% higher than that registered in the reactor without aeration.

Tomei et al. [60] studied the hygienization performance for sewage sludge under a sequential anaerobic–aerobic process. The study showed that implementation of an aerobic phase after anaerobic digestion may reduce *E.coli* population, a conventional microbial indicator originating from human and animal intestines. An increase in temperature during the aerobic stage was found to have no effect on lowering *E.coli* population.

It should however be underlined that to date little is known about the biological stability of digestate subjected to a micro-aerobic AD process. Previous studies have been strictly divided into anaerobic and aerobic processes with aeration being applied as post-treatment (composting) to achieve the stability of digestate [61–63]. However, it is possible to improve both methane yield and digestate biostability by combining anaerobic/aerobic treatment in the same AD process [11, 15, 46]. Further research should be

Table 4 Effects of limited aeration during the AD process on the H_2S reduction

Substrate	AD process and test scale	Aeration method and amount	Results	References
Sewage sludge	Mesophilic CSTR (200 L)	Oxygen supply 0–0.00485 $LO_2/L_R/d$ for low H_2S concentration; 0–0.0705 LO_2/d for high H_2S concentration	97% H_2S removal efficiency at low concentration (0.33% in the biogas) and 99% at high concentration (3.38% H_2S in the biogas)	[56]
Sewage sludge	Mesophilic CSTR (265 L)	Oxygen supply from a cylinder by a mass flow controller 0.16–0.46 LO_2/L_{fed}	99% H_2S removal efficiency applying 0.46 LO_2/L_{fed}	[57]
Sludge from WWTP	Mesophilic CSTR (200 L)	Oxygen injection 0.013–0.024 $LO_2/L_R/d$	More than 99% H_2S removal	[54]
Synthetic brewery wastewater	UASB (2.7 L)	Air flow 0.37 $L/L_R/d$	73% H_2S removal efficiency	[58]
Thickened surplus activated sludge	Mesophilic CSTR (11 L)	Air pump 1.1 Lair/d (0.1 $L/L_R/d$)	92.3% H_2S reduction	[59]

conducted on the aerobic treatment in the late AD stage, with particular focus on easier digestate stability.

Effect of Aeration on Microbial Diversity

Information relating to the presence of microbial communities in AD reactors operating under strictly anaerobic and limited aeration conditions is crucial to provide a better insight into the metabolic reactions involved in the process. The availability of this information promotes an easier and better control of the AD process parameters, and contributes significantly towards improving process performance. However, very few reports focus on the presence of microbial communities following limited aeration. Micro-aeration may potentially shorten the lag phase, increase the specific growth rate and improve tolerability of microorganisms to substrate inhibition [64]. Moreover, micro-aeration leads to a more diverse microbial variety, a crucial issue for deep substrate decomposition. An interesting study revealed that the presence of *Acetobacterperoxydans*, an aerobic bacterium, led to high VFA production inside the acidogenic reactor [7]. Tang et al. [65] investigated the effect of micro-aeration on thermophilic anaerobic digestion of municipal solid waste (MSW), with no significant difference in CH_4 production being observed between micro-aerobic and anaerobic conditions. Although no significant difference was observed in microbial diversity under micro-aeration conditions, however, the number of Methanosarcina decreased while Methanoculleus population increased. This was mainly due to the difference in oxygen tolerability of the two different methanogenic archaeal species. Although micro-aeration did not suppress the activity of sulphate-reducing bacteria (SRB), H_2S production under micro-aeration was much lower. The Authors concluded that limited oxygen chemically improved sulphide oxidation in partially aerated reactors. In contrast to Tang et al. [65], Ramos et al. [43] did not observe any significant impact on archaeal populations following micro-aeration.

Another study conducted to investigate AD treatment in a sulphite pulp mill showed how limited aeration is capable of reducing sulphide inhibition of methanogens, highlighting how inhibition decreased gradually in the presence of aeration due to an increase in sulphide oxidation. Predominant microorganisms present before aeration were rod-shape methanogens, while after limited aeration cocci-shaped methanogens were dominant [6].

A study carried out by Lim et al. [44] under conditions of micro-aeration revealed a greater diversity in the microbial population compared to that observed under strictly anaerobic conditions. This diversity led to improved hydrolysis rates and subsequently better process performance.

Conclusions

Aerobic treatment exerts both positive and negative effects when combined with AD. The overall influence reported in the literature does not allow any definitive conclusions to be drawn. The effect produced is mainly associated with duration of specific aeration, intensity, and methods used (air or oxygen only, injected or pumped in through intermittent or continuous mode), on the air application phase (before, during, or at a late stage of AD), on the type of AD test (batch reactors or CSTR), and on the type of digested substrate (e.g. corn straw, sewage sludge, food waste). Numerous reports have referred to an enhanced methane production during AD following the application of aeration. Aerobic treatment is capable of removing hydrogen sulphides that accumulate during the AD of sulphur-rich substrates; micro-aeration improves hydrolysis of recalcitrant organic matters such as corn straw; moreover, the problem of acidification when starting AD operations in batch reactors can be prevented by pre-aeration of organic matter such as OFMSW.

Aerobic treatment may be capable not only of improving the performance of AD, but also of ensuring an easier post-treatment of digestate. However, some Authors have obtained lower methane yields after aerobic pre-treatment of the digested substrates or a poorer biogas quality after aeration during AD.

The conducting of microbial diversity studies would likely be beneficial to investigate the effect produced by a series of substrates and aeration conditions on different methanogenic microorganisms. Ecotoxicological tests considering the potential pollutants contained in the digestate are suggested as a further tool to assess environmental benefits of using aeration in combination with anaerobic digestion processes [66–68].

Use of a combination of aerobic treatment with other pre-treatment methods (e.g. thermal chemical pre-treatment, inoculum recirculation) should be taken into consideration in an attempt to achieve a synergistic effect. Several novel aeration methods, such as electrolytic aeration, should also be further investigated.

Economic feasibility studies should likewise be undertaken to examine additional aeration costs [42]; this parameter, together with the demonstration of improved AD performance, should represent the key tool in the decision-making process.

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